

1. Administrative & Study Identification

Full Study Title: A Phase 2 Multi-Cohort, Open-Label, Multi-Center Clinical Study Evaluating the Efficacy and Safety of Disitamab Vedotin (RC48-ADC) Alone or in Combination With Pembrolizumab in Subjects With Locally-Advanced Unresectable or Metastatic Urothelial Carcinoma That Expresses HER2 **Short Title/ Acronym:** A Study of Disitamab Vedotin Alone or With Pembrolizumab in Urothelial Cancer That Expresses HER2 (RC48G001 / KEYNOTE-D78) **Protocol Number:** RC48G001 / C5731002 **NCT Number:** NCT04879329 **Posting ID:** 48603868157989883 **Sponsor Name:** Pfizer **Investigational Product:** Disitamab vedotin (RC48-ADC) alone or in combination with pembrolizumab (Keytruda) **Phase of Development:** Phase II **Trial Status:** RECRUITING **Medical Monitor:** Peter H. O'Donnell / Xuemei Li (xuli@seagen.com)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy (specifically the confirmed objective response rate per blinded independent central review) and safety (incidence of adverse events and dose alterations) of disitamab vedotin alone or in combination with pembrolizumab in patients with HER2-expressing locally advanced/metastatic urothelial carcinoma across multiple patient cohorts. **Secondary Objectives:** To evaluate investigator-assessed cORR, Duration of Response (DOR) by both BICR and investigator assessment, Progression-Free Survival (PFS), Overall Survival (OS), Disease Control Rate (DCR), and additional safety and tolerability metrics of the study drugs.

2.2 Background & Rationale To address the significant unmet medical need in patients with locally advanced unresectable or metastatic urothelial carcinoma (LA/mUC) whose tumors express HER2. Nearly 50% of patients cannot tolerate current first-line standards of care, and despite HER2 overexpression in UC, no HER2-directed therapies are currently approved. Disitamab vedotin, a novel HER2-directed antibody-drug conjugate (ADC) conjugated to monomethyl auristatin E (MMAE), may provide robust antitumor activity through targeted cytotoxicity and immunogenic cell death, which can be further enhanced when combined with the PD-1 inhibitor pembrolizumab.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm / Multi-cohort **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$\$\$: Approximately 372 evaluable patients **Number of Sites**: Multi-center (Global, including North America, Europe, Latin America, and Asia-Pacific) **Estimated Study Start**: Mid-2021 **Estimated Primary Completion**: Q3-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histopathologically-confirmed, locally-advanced, unresectable or metastatic urothelial cancer (LA/mUC), including UC originating from the renal pelvis, ureters, bladder, or urethra. 2. HER2-expression status determined by the central laboratory to be IHC 1+, 2+, or 3+ in the provided tumor sample. 3. At least one measurable lesion by investigator assessment based on RECIST version 1.1. 4. Prior systemic therapy requirements vary by cohort: * **Cohorts A & B**: 1 or 2 lines of prior systemic treatment, including 1 line of platinum-containing chemotherapy. * **Cohorts C & E**: No prior systemic therapy for LA/mUC (eligible for cisplatin/carboplatin). * **Cohort D**: Must have received a platinum-containing chemotherapy, a PD-(L)1 inhibitor, and enfortumab vedotin. * **Cohort G**: 1 or 2 lines of prior systemic treatment, including 1 line containing enfortumab vedotin.

4.2 Exclusion Criteria 1. Known hypersensitivity to disitamab vedotin, pembrolizumab, or any of their components. 2. Prior antitumor treatment within 2 weeks of the start of study treatment. 3. Prior treatment with MMAE-based ADCs (e.g., enfortumab vedotin) or HER2-directed therapy (unless expressly permitted by a specific sub-cohort like Cohort D or G). 4. Peripheral sensory or motor neuropathy $\geq$ Grade 2 at baseline (Cohorts A, B, C, G) or $\geq$ Grade 1 at baseline (Cohorts D, E). 5. Immunodeficiency or receiving chronic systemic steroid therapy within 7 days prior to the first dose (Cohorts C, E).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Disitamab vedotin administered once every 2 weeks (Q2W). For combination cohorts, pembrolizumab (Keytruda) is administered on Day 1 of each 6-week cycle. **Route of Administration**: Intravenous (IV) infusion **Duration of Treatment**: Until disease progression, unacceptable toxicity, or completion of approximately 2 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for symptom management. **Prohibited**: Prior antitumor treatment (including chemotherapy, radiotherapy, targeted therapy, immunotherapy) within 2 weeks of the start of study treatment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor biopsy for Central HER2 testing, Baseline Imaging (RECIST v1.1)
Baseline	Day 1	Eligibility Confirmation, First Dose of Disitamab Vedotin (+/- Pembrolizumab)
Interim	Every 2 to 6 Weeks	Safety Labs, IV Dosing, Efficacy Assessment (Scans), AE monitoring
End of Study	Up to ~Year 2	Final Scans, Exit Interview, IP Accountability, Survival Follow-up Phase initiation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- Confirmed Objective Response Rate (cORR)** per Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1 by blinded independent central review (BICR) (Cohorts A, B, C, and G).
- Incidence of adverse events (AEs)** (Cohorts D and E).
- Incidence of dose alterations** (Cohorts D and E).

7.2 Secondary & Exploratory Endpoints

- * cORR per RECIST v1.1 by investigator assessment (Cohorts A, B, C, and G).
- * **Confirmed Duration of Response (DOR)** per RECIST v1.1 by BICR (Cohorts A, B, C, and G).
- * **Confirmed DOR** per RECIST v1.1 by investigator assessment (Cohorts A, B, C, and G).
- * Progression-Free Survival (PFS)
- * Overall Survival (OS)
- * Disease Control Rate (DCR)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via clinical evaluations, laboratory assessments, and monitoring for specific AEs of interest (e.g., fatigue, anemia, peripheral neuropathy, and potential pneumonitis). Specific focus on incidence of AEs and dose alterations in Cohorts D and E.

Serious Adverse Events (SAEs): Reported within 24 hours of site awareness.

Data Monitoring Committee (DMC): Independent Data Monitoring Committee to oversee patient safety and interim safety run-in data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all AE/SAE and tolerability evaluations.

Sample Size Justification: Approximately 372 participants, designed to provide sufficient statistical power ($\alpha = 0.05$) to evaluate ORR across distinct HER2-expressing cohorts and treatment lines with precision.

Handling of Missing Data: Standard oncology guidelines; censoring applied for time-to-event endpoints (PFS, OS, DOR) at the last evaluable assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits to ensure protocol adherence and patient safety. **Audit Trail:**

Continuous eCRF tracking, secure centralized pathology review for HER2 status, and blinded independent central review (BICR) for imaging.

11. Study Identifiers & Governance

Primary ID: RC48G001 / C5731002 **Registry Number:** NCT04879329

Posting ID: 48603868157989883 **Sponsor/Lead Organization:** Pfizer

Study Title: A Study of Disitamab Vedotin Alone or With Pembrolizumab in Urothelial Cancer That Expresses HER2 **Official**

Regulatory Title: A Phase 2 Multi-Cohort, Open-Label, Multi-Center Clinical Study Evaluating the Efficacy and Safety of Disitamab Vedotin (RC48-ADC) Alone or in Combination With Pembrolizumab in Subjects With Locally-Advanced Unresectable or Metastatic Urothelial Carcinoma That Expresses HER2

12. Clinical Framework (Urothelial Carcinoma Specific)

Primary Condition: Locally-Advanced Unresectable or Metastatic Urothelial Carcinoma (LA/mUC) **Disease Area:** Urothelial Carcinoma

Preferred UMLS Name: HER2 Positive Urothelial Carcinoma **Diagnosis**

Threshold: Histopathological confirmation and tumor HER2-expression of IHC 1+, 2+, or 3+ **Medical Methodology:** HER2-directed Antibody-Drug Conjugate (ADC) combined with PD-1 checkpoint inhibition **Optimization Target:** Maximal tumor regression (Complete/Partial Response) and prolonged Progression-Free Survival

13. Study Procedures & Methodology

Management Pathway: Protocol-directed systemic antineoplastic therapy with standardized imaging intervals. **Education Protocol**

(HCP): Training on protocol guidelines, specifically early identification and management of ADC-related toxicities (e.g., neuropathy, hematologic toxicity). **Education Protocol (Patient):**

Informed consent process, review of treatment schedule, and self-monitoring for side effects. **Quality Audit Mechanism:** Independent central pathology for HER2 status confirmation and blinded independent central review (BICR) for RECIST v1.1 tumor measurements.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 2 years of active treatment, followed by survival follow-up. **Onsite Visit Cadence:** Every 2 weeks for Disitamab vedotin administration; Every 6 weeks for Pembrolizumab administration. **Remote Monitoring Frequency:** Routine safety check-ins as required per standard oncology protocols. **Checklist Review Interval:** Routine radiologic imaging for disease assessment typically conducted every 6 to 9 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				